



Parul University

FACULTY OF ENGINEERING AND TECHNOLOGY
BACHELOR OF TECHNOLOGY

Mobile App Development Laboratory
(303105380)

6th SEMESTER (3RD YEAR)
COMPUTER SCIENCE & ENGINEERING DEPARTMENT

LAB MANUAL



CERTIFICATE

This is to certify that Mr./Ms **KUNAM MANI SAI CHARAN REDDY** with Enrollment no. **2303031250067** has successfully completed his/her laboratory experiments in **Mobile App Development Laboratory (303105380)** from the Department of Computer Science and Engineering (Big Data Analytics) during the academic year 2025-26.

Date of Submission:

Staff In charge: Head of Department:

INDEX

Sr. no	Practical	Page no.	Date of Assessment	Date of Submission	Mark	Sign
1	Create a "Hello World" application					
2	Build an app to showcase Android lifecycle phases					
3	Create an app with two activities					
4	Create an app with explicit intent					
5	Build a basic calculator app.					
6	Create a Spinner-based app.					
7	Create a discount calculator app.					
8	Create an app with a course selection Radio Button group.					
9	Create a shopping list app using checkboxes.					
10	Create a login and registration app					

11	Create a login app with navigation to another activity:					
12	Create an app with a menu:					
13	Build an app using Linear Layout.					
14	Create an app with Table Layout and custom styles.					
15	Create an app with SQLite database operations					
16	Create an app with three vertically aligned buttons					

Practical-1

Aim: Create a "Hello World" application

Xml file code(activity_main.xml):

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World 2"/>

</LinearLayout>
```

MainActivity.java code:

```
package com.example.myapplication;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
```

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);
    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
        Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
        v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
        return insets;
    });
}
```

Output:



Practical-2

Aim: Create an app with explicit intent

activity_main.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Retro Phone Booth"
        android:textAlignment="center"
        android:textStyle="bold"
        android:background="#a25654"
        android:textColor="#fff"
        android:textSize="40dp"/>

    <ImageView
        android:layout_width="wrap_content"
        android:layout_height="300dp"
        android:src="@drawable/tell" />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:paddingRight="10dp"
        android:paddingLeft="10dp">

        <EditText
            android:id="@+id/eText"
            android:layout_width="match_parent"
            android:paddingRight="10dp"
            android:paddingLeft="10dp"
            android:layout_height="50dp"
            android:layout_gravity="center"
            android:hint="Enter Your Number"/>

        <Button
            android:id="@+id/callBtn"
```

```
android:layout_width="120dp"
android:layout_height="50dp"
android:text="Call Now"
android:layout_marginTop="10dp"
android:layout_gravity="center"
android:backgroundTint="#a25654"/>
```

```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="bottom"
    android:orientation="vertical">
```

```
<com.google.android.material.button.MaterialButton
    android:id="@+id/shareBtn"
    android:layout_width="100dp"
    android:layout_height="60dp"
    app:icon="@drawable/share_icon"
    app:iconSize="28dp"
    app:iconGravity="textEnd"
    android:contentDescription="share"
    android:layout_gravity="end"
    android:layout_marginBottom="20dp"
    android:backgroundTint="#a25654"/>
```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
</LinearLayout>
```

MainActivity.java code:

```
package com.example.implicit_intent;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```



```
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        //----- creating the button -----
        EditText editText;
        Button callBtn, shareBtn;

        //----- finding that view by id -----
        editText= findViewById(R.id.eText);
        callBtn = findViewById(R.id.callBtn);
        shareBtn = findViewById(R.id.shareBtn);

        //----- creating intent -----
        //this intent is for dial number
        Intent dialIntent = new Intent(Intent.ACTION_DIAL);
        Intent shareIntent = new Intent(Intent.ACTION_SEND);

        //----- code for call button click -----
        callBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                //---- extracting text from input field -----
                String number = editText.getText().toString();
                dialIntent.setData(Uri.parse("tel: "+ number));
                startActivity(dialIntent);
            }
        });

        // ----- code for share button click -----
        shareBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                shareIntent.setType("text/plain");
                shareIntent.putExtra(Intent.EXTRA_TEXT, "My Portfolio,
https://saketfolio.netlify.app/");
                startActivity(shareIntent);
            }
        });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
        {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
```

```
v.setPadding(systemBars.left, systemBars.top,  
systemBars.right, systemBars.bottom);  
return insets;  
});  
}  
}
```

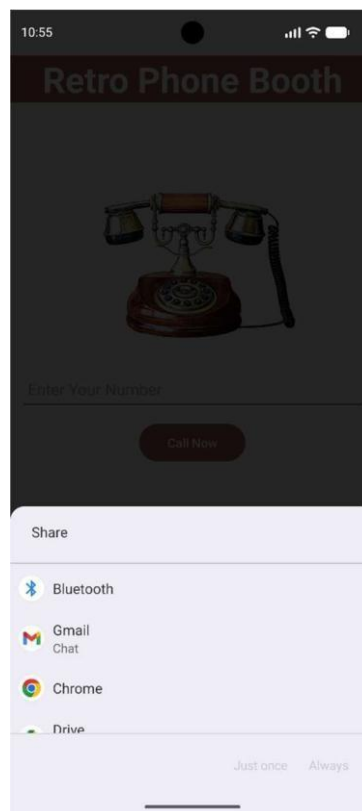
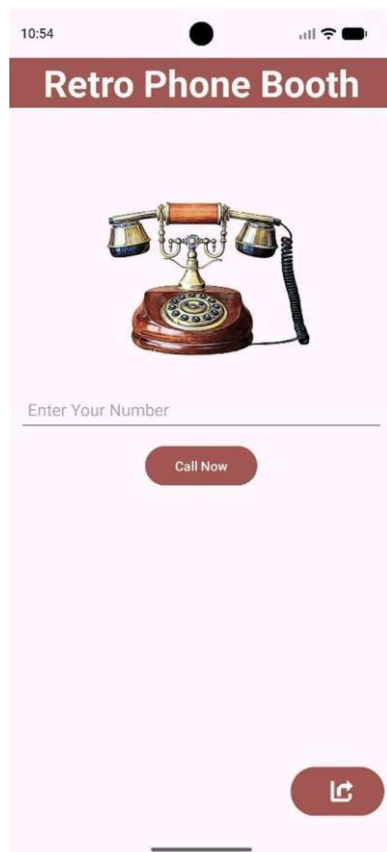
colors.xml code:

```
<?xml version="1.0" encoding="utf-8"?>  
<resources>  
    <color name="black">#FF000000</color>  
    <color name="white">#FFFFFFF</color>  
</resources>
```

strings.xml code:

```
<resources>  
    <string name="app_name">Implicit_Intent</string>  
</resources>
```

Output:



Practical-3

Aim: Create an app with two activities

activity_main.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:paddingLeft="12dp"
    android:paddingRight="12dp"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="First Screen"
        android:textSize="30dp"
        android:textAlignment="center"
        android:textStyle="bold"/>

    <ImageView
        android:layout_width="match_parent"
        android:layout_height="300dp"
        android:src="@drawable/tom"/>

    <!-- <ImageView
        android:layout_width="match_parent"
        android:layout_height="300dp"
        android:src="@drawable/tom"/>-->

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="We Are Friends"
        android:textSize="20dp"
        android:textAlignment="center"
        android:layout_marginBottom="20dp"
        android:textStyle="bold"/>

    <EditText
        android:id="@+id/eText"
        android:layout_width="match_parent"
        android:layout_height="50dp"
```

```
android:textColor="@color/black"  
android:textSize="20dp"  
android:hint="Enter Your Name"/>
```

```
<LinearLayout  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:gravity="bottom|center"  
    android:orientation="vertical">  
  
    <Button  
        android:id="@+id/nextBtn"  
        android:layout_width="120dp"  
        android:layout_height="60dp"  
        android:textStyle="bold"  
        android:textSize="18dp"  
        android:backgroundTint="#ff0000"  
        android:layout_marginBottom="100dp"  
        android:text="Next"/>  
</LinearLayout>
```

```
</LinearLayout>
```

MainActivity.java

```
package com.example.intent;  
  
import android.content.Intent;  
import android.os.Bundle;  
import android.view.View;  
import android.widget.Button;  
import android.widget.EditText;  
  
import androidx.activity.EdgeToEdge;  
import androidx.appcompat.app.AppCompatActivity;  
import androidx.core.graphics.Insets;  
import androidx.core.view.ViewCompat;  
import androidx.core.view.WindowInsetsCompat;  
  
public class MainActivity extends AppCompatActivity {  
  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        EdgeToEdge.enable(this);  
        setContentView(R.layout.activity_main);  
  
        // calling button  
        Button nextBtn;  
        nextBtn = findViewById(R.id.nextBtn);
```

```
EditText eText;
eText = findViewById(R.id.eText);

// making intent
Intent inext;

// use inbuilt class to navigate btwn activity
inext = new Intent(MainActivity.this , SecondActivity.class);

//----- to share data from main act to second act -----
// adding click method to the button

nextBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

// passing the data as string and int
        String myname = eText.getText().toString();
        inext.putExtra("myName", myname);

        inext.putExtra("myName", "Shubhangi Jha");
// start activity of the intent named as inext
        startActivity(inext);

    }
});

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
{
    Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(
        systemBars.left + v.getPaddingLeft(),
        systemBars.top + v.getPaddingTop(),
        systemBars.right + v.getPaddingRight(),
        systemBars.bottom + v.getPaddingBottom()
    );
    return insets;
});
}
```

activity_second.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".SecondActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Second Screen"
        android:textSize="30dp"
        android:textStyle="bold"
        android:textAlignment="center"/>

    <TextView
        android:id="@+id/tName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Name: "
        android:textSize="25dp"
        android:textStyle="bold"
        android:layout_marginTop="20dp"
        android:textAlignment="center"/>

    <!--<TextView
        android:id="@+id/tAge"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Age: "
        android:textSize="25dp"
        android:textStyle="bold"
        android:layout_marginTop="20dp"
        android:textAlignment="center"/>
-->

</LinearLayout>
```

SecondActivity.java code

```
package com.example.intent;

import android.content.Intent;
import android.os.Bundle;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class SecondActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_second);

        // getting intent from main-activity
        Intent fromMain = getIntent();

        // storing the values in variable
        String name = fromMain.getStringExtra("myName").toString();
        // String sName = fromMain.getStringExtra("secondName").toString();
        // int age = fromMain.getIntExtra("age", 0);

        // calling text-view as id name
        TextView tName;
        TextView sName;
        //TextView tAge;

        // finding the view by id
        // tAge = findViewById(R.id.tAge);
        tName = findViewById(R.id.tName);

        // set this data in view
        tName.setText(name);
        //tAge.setText(age + "");

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
        {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

```
}  
}
```

activity_third.xml code:

```
<?xml version="1.0" encoding="utf-8"?>  
<androidx.constraintlayout.widget.ConstraintLayout  
    xmlns:android="http://schemas.android.com/apk/res/android"  
        xmlns:app="http://schemas.android.com/apk/res-auto"  
        xmlns:tools="http://schemas.android.com/tools"  
        android:id="@+id/main"  
        android:layout_width="match_parent"  
        android:layout_height="match_parent"  
        tools:context=".ThirdActivity">  
  
</androidx.constraintlayout.widget.ConstraintLayout>
```

ThirdActivity.java

```
package com.example.intent;  
  
import android.os.Bundle;  
  
import androidx.activity.EdgeToEdge;  
import androidx.appcompat.app.AppCompatActivity;  
import androidx.core.graphics.Insets;  
import androidx.core.view.ViewCompat;  
import androidx.core.view.WindowInsetsCompat;  
  
public class ThirdActivity extends AppCompatActivity {  
  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        EdgeToEdge.enable(this);  
        setContentView(R.layout.activity_third);  
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->  
        {  
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());  
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);  
            return insets;  
        });  
    }  
}
```

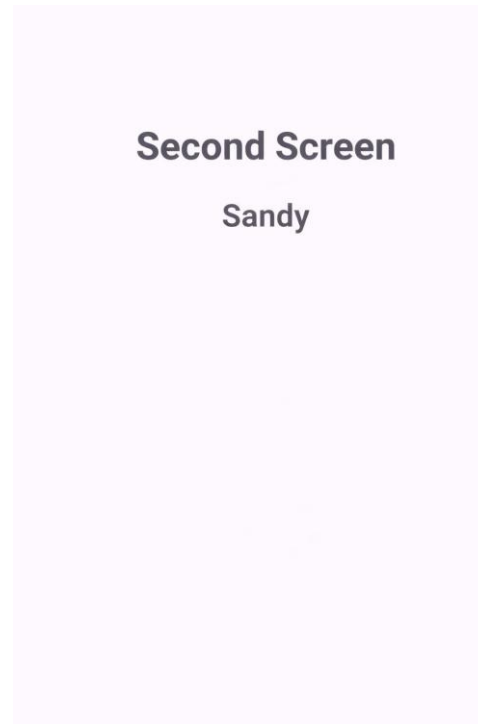
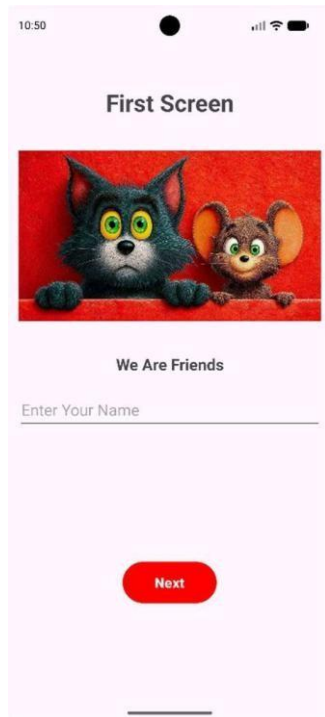
colors.xml code:


```
<?xml version="1.0" encoding="utf-8"?>
<resources>
  <color name="black">#FF000000</color>
  <color name="white">#FFFFFFF</color>
</resources>
```

strings.xml code:

```
<resources>
  <string name="app_name">Intent</string>
</resources>
```

OUTPUT



Practical-4

Aim: Create an app with explicit intent

activity_main.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Retro Phone Booth"
        android:textAlignment="center"
        android:textStyle="bold"
        android:background="#a25654"
        android:textColor="#fff"
        android:textSize="40dp"/>

    <ImageView
        android:layout_width="wrap_content"
        android:layout_height="300dp"
        android:src="@drawable/tell" />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:paddingRight="10dp"
        android:paddingLeft="10dp">

        <EditText
            android:id="@+id/eText"
            android:layout_width="match_parent"
            android:paddingRight="10dp"
            android:paddingLeft="10dp"
            android:layout_height="50dp"
            android:layout_gravity="center"
            android:hint="Enter Your Number"/>

        <Button
```

```
android:id="@+id/callBtn"  
android:layout_width="120dp"  
android:layout_height="50dp"  
android:text="Call Now"  
android:layout_marginTop="10dp"  
android:layout_gravity="center"  
android:backgroundTint="#a25654"/>
```

```
<LinearLayout  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:gravity="bottom"  
    android:orientation="vertical">
```

```
<com.google.android.material.button.MaterialButton  
    android:id="@+id/shareBtn"  
    android:layout_width="100dp"  
    android:layout_height="60dp"  
    app:icon="@drawable/share_icon"  
    app:iconSize="28dp"  
    app:iconGravity="textEnd"  
    android:contentDescription="share"  
    android:layout_gravity="end"  
    android:layout_marginBottom="20dp"  
    android:backgroundTint="#a25654"/>
```

```
</LinearLayout>
```

```
</LinearLayout>
```

```
</LinearLayout>
```

MainActivity.java code:

```
package com.example.implicit_intent;  
  
import android.content.Intent;  
import android.net.Uri;  
import android.os.Bundle;  
import android.view.View;  
import android.widget.Button;  
import android.widget.EditText;  
  
import androidx.activity.EdgeToEdge;  
import androidx.appcompat.app.AppCompatActivity;  
import androidx.core.graphics.Insets;  
import androidx.core.view.ViewCompat;
```

```
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        //----- creating the button -----
        EditText editText;
        Button callBtn, shareBtn;

        //----- finding that view by id -----
        editText= findViewById(R.id.eText);
        callBtn = findViewById(R.id.callBtn);
        shareBtn = findViewById(R.id.shareBtn);

        //----- creating intent -----
        //this intent is for dial number
        Intent dialIntent = new Intent(Intent.ACTION_DIAL);
        Intent shareIntent = new Intent(Intent.ACTION_SEND);

        //----- code for call button click -----
        callBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                //---- extracting text from input field -----
                String number = editText.getText().toString();
                dialIntent.setData(Uri.parse("tel: "+ number));
                startActivity(dialIntent);
            }
        });

        // ----- code for share button click -----
        shareBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                shareIntent.setType("text/plain");
                shareIntent.putExtra(Intent.EXTRA_TEXT, "My Portfolio,
https://saketfolio.netlify.app/");
                startActivity(shareIntent);
            }
        });

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
        {
```

```
Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
    return insets;
    });
    }
}
```

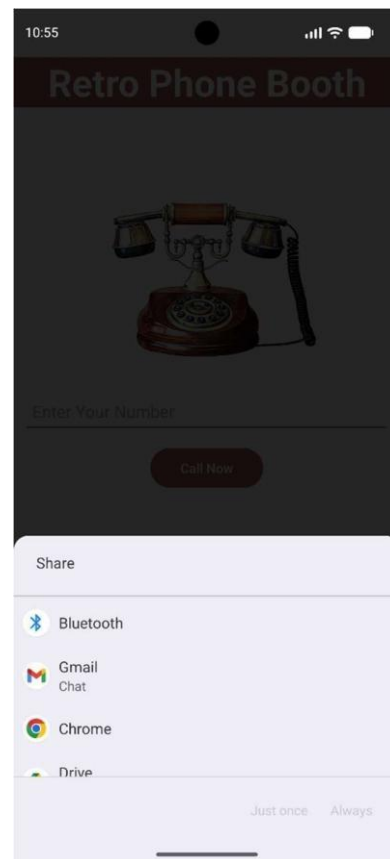
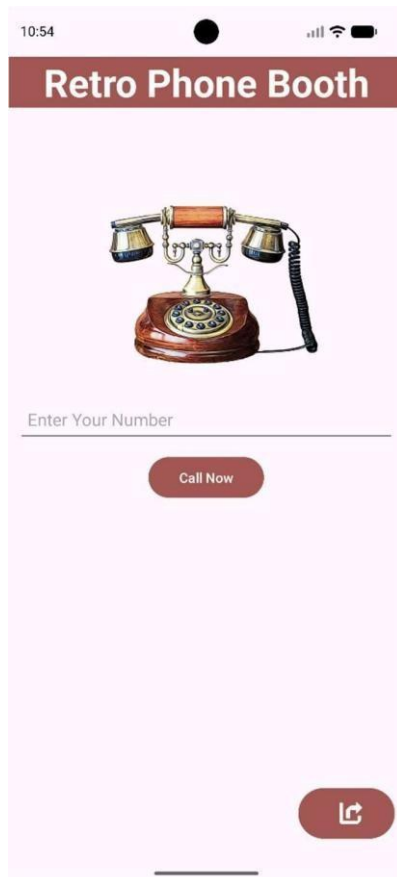
colors.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
</resources>
```

strings.xml code:

```
<resources>
    <string name="app_name">Implicit_Intent</string>
</resources>
```

Output:



Practical-5

Aim: Build a basic calculator app.

Activity_main.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    android:orientation="vertical"
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="CALCULATOR"
        android:textSize="48sp"
        android:textAlignment="center"
        android:textStyle="bold"
        android:fontFamily="@font/sophiefancy"
        android:layout_marginTop="20dp"
        android:letterSpacing="0.07"
        />

    <EditText
        android:id="@+id/firstValue"
        style="@style/editText"
        android:hint="Enter 1st Value"
        android:layout_marginTop="100dp"
        />

    <EditText
        android:id="@+id/secValue"
        android:hint="Enter 2nd Value"
        style="@style/editText"
        android:layout_marginTop="60dp"
        />

    <TextView
```

```
android:id="@+id/ans"  
android:layout_width="match_parent"  
android:layout_height="wrap_content"  
android:text="Result: "  
android:fontFamily="@font/sophiefancy"  
android:textAlignment="center"  
android:textSize="30sp"  
android:textStyle="bold"  
android:layout_marginTop="50dp"  
</>
```

```
<LinearLayout  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:layout_marginTop="80dp"  
    android:padding="10dp"  
    android:orientation="horizontal">
```

```
<Button  
    android:id="@+id/addBtn"  
    style="@style/buttonView"  
    android:text="+" />
```

```
<Button  
    android:id="@+id/subBtn"  
    style="@style/buttonView"  
    android:text="-" />
```

```
</LinearLayout>
```

```
<LinearLayout  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:layout_marginTop="10dp"  
    android:padding="10dp"  
    android:orientation="horizontal">
```

```
<Button  
    android:id="@+id/mulBtn"  
    style="@style/buttonView"  
    android:text="*" />
```

```
<Button  
    android:id="@+id/divBtn"  
    style="@style/buttonView"  
    android:text="/" />
```

```
</LinearLayout>
```

```
<TextView  
    android:layout_width="match_parent"
```

```
android:layout_height="wrap_content"
android:text="@string/developer "
android:textAlignment="center"
android:layout_marginTop="30dp"
android:textSize="20dp"
android:fontFamily="@font/sophiefancy"
android:layout_gravity="end"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.calculator;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        // creating edit text
        TextView ans;
        EditText firstValue, secValue;

        // finding that edit text
        ans = findViewById(R.id.ans);
        firstValue = findViewById(R.id.firstValue);
        secValue = findViewById(R.id.secValue);

        // creating button
        Button addBtn, subBtn, mulBtn, divBtn;

        // finding these buttons
        addBtn = findViewById(R.id.addBtn);
        subBtn = findViewById(R.id.subBtn);
        mulBtn = findViewById(R.id.mulBtn);
```



```
divBtn = findViewById(R.id.divBtn);

// code for add button click
addBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        String value1 = firstValue.getText().toString();
        String value2 = secValue.getText().toString();

        if(value1.isEmpty() || value2.isEmpty()){
            Toast.makeText(MainActivity.this, "Enter Value",
Toast.LENGTH_SHORT).show();
            ans.setText("Result: ");
            return;
        }else {
            int num1 = Integer.parseInt(value1);
            int num2 = Integer.parseInt(value2);

            int result = num1 + num2;
            ans.setText("Result: "+ result);
        }
    }
});

// code for sub button click
subBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        String value1 = firstValue.getText().toString();
        String value2 = secValue.getText().toString();

        if(value1.isEmpty() || value2.isEmpty()){
            Toast.makeText(MainActivity.this, "Enter Value",
Toast.LENGTH_SHORT).show();
            ans.setText("Result: ");
            return;
        }else {
            int num1 = Integer.parseInt(value1);
            int num2 = Integer.parseInt(value2);

            int result = num1 - num2;
            ans.setText("Result: "+ result);
        }
    }
});

// code for mul button click
mulBtn.setOnClickListener(new View.OnClickListener() {
```

```
@Override
public void onClick(View v) {
    String value1 = firstValue.getText().toString();
    String value2 = secValue.getText().toString();

    if(value1.isEmpty() || value2.isEmpty()){
        Toast.makeText(MainActivity.this, "Enter Value",
Toast.LENGTH_SHORT).show();
        ans.setText("Result: ");
        return;
    }else {
        int num1 = Integer.parseInt(value1);
        int num2 = Integer.parseInt(value2);

        int result = num1 * num2;
        ans.setText("Result: "+ result);
    }
}

});
```

```
// code for div button click
divBtn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        String value1 = firstValue.getText().toString();
        String value2 = firstValue.getText().toString();

        if(value1.isEmpty() || value2.isEmpty()){
            Toast.makeText(MainActivity.this, "Enter Value",
Toast.LENGTH_SHORT).show();
            ans.setText("Result: ");
            return;
        }else {
            int num1 = Integer.parseInt(value1);
            int num2 = Integer.parseInt(value2);

            int result = num1 / num2;
            ans.setText("Result: "+ result);
        }
    }
});
```

```
ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
{
```

```
Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
    return insets;
    });
    }
}
```

colors.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFF</color>
</resources>
```

Strings.xml code:

```
<resources>
    <string name="app_name">Calculator</string>
    <string name="developer">Developed by @ Shubhangi Jha</string>
</resources>
```

Styles.xml code:

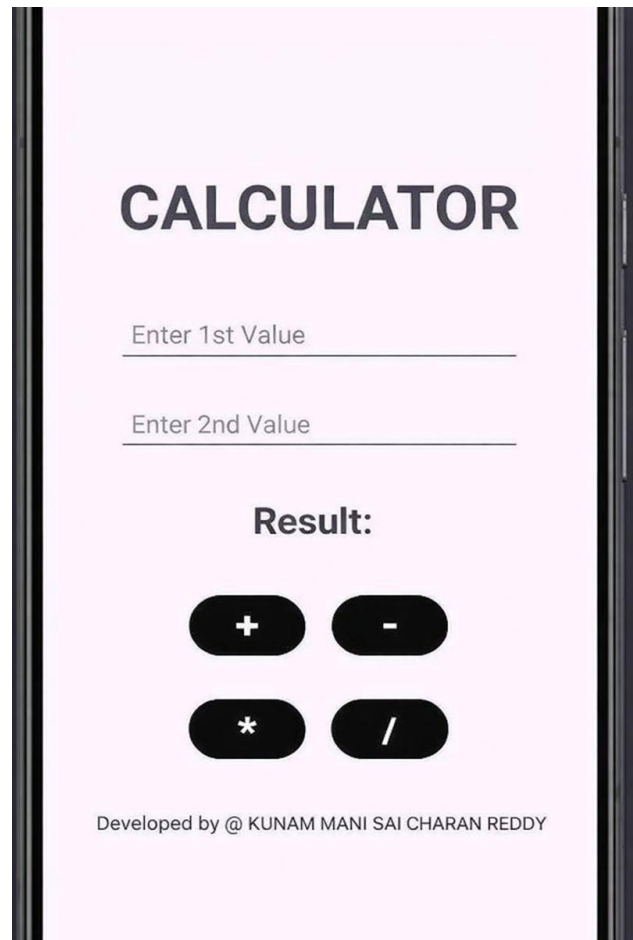
```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    // style for edit text view

    <style name="editText">
        <item name="android:textSize">20dp</item>
        <item name="android:layout_width">300dp</item>
        <item name="android:padding">10dp</item>
        <item name="android:layout_gravity">center</item>
        <item name="android:layout_height">50dp</item>
        <item name="android:inputType">number</item>
    </style>

    // style for button view

    <style name="buttonView">
        <item name="android:layout_height">wrap_content</item>
        <item name="android:layout_width">wrap_content</item>
        <item name="android:textSize">30sp</item>
        <item name="android:layout_weight">1</item>
        <item name="android:textStyle">bold</item>
        <item name="backgroundTint">@color/black</item>
        <item name="android:layout_marginEnd">10dp</item>
    </style>
</resources>
```

Output:



Practical-6

Aim: Create a Spinner-based app.

Activity_main.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:orientation="vertical"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="10dp"
    android:gravity="center"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="AVENGERS ASSEMBLE"
        android:textSize="32sp"
        android:textAlignment="center"
        android:textStyle="bold"
        android:layout_marginTop="20dp"
    />

    <Spinner
        android:id="@+id/spinner"
        android:layout_gravity="center"
        android:layout_width="300dp"
        android:layout_height="50dp"
        android:layout_marginTop="40dp"
        android:textAlignment="center"
        android:padding="10dp"
    />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:gravity="bottom">

        <ImageView
            android:id="@+id/imageview"
            android:layout_width="400dp"
```

```
android:layout_height="400dp"  
android:layout_gravity="center"  
android:src="@drawable/caption"  
android:layout_marginBottom="30dp"  
>
```

```
</LinearLayout>
```

```
</LinearLayout>
```

Avenger_text.xml code:

```
<?xml version="1.0" encoding="utf-8"?>  
<TextView xmlns:android="http://schemas.android.com/apk/res/android"  
    android:id="@+id/text"  
    android:layout_width="wrap_content"  
    android:layout_height="50dp"  
    android:textSize="22dp"  
    android:padding="10dp"  
    android:textColor="@color/black">  
</TextView>
```

Colors.xml code:

```
<?xml version="1.0" encoding="utf-8"?>  
<resources>  
    <color name="black">#FF000000</color>  
    <color name="white">#FFFFFFFF</color>  
</resources>
```

Strings.xml code:

```
<resources>  
    <string name="app_name">Spinner</string>  
  
    <string-array name="avengers">  
        <item>Caption</item>  
        <item>Iron Man</item>  
        <item>Black Panther</item>  
        <item>Hulk</item>  
        <item>Thor</item>  
    </string-array>  
  
</resources>
```

```
package com.example.spinner;

import android.media.Image;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ImageView;
import android.widget.Spinner;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    int[] images = {R.drawable.caption, R.drawable.ironman, R.drawable.blackpanther,
        R.drawable.hulk,
            R.drawable.thor};

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        Spinner spinner;
        spinner = findViewById(R.id.spinner);

        ImageView imageview;
        imageview = findViewById(R.id.imageview);

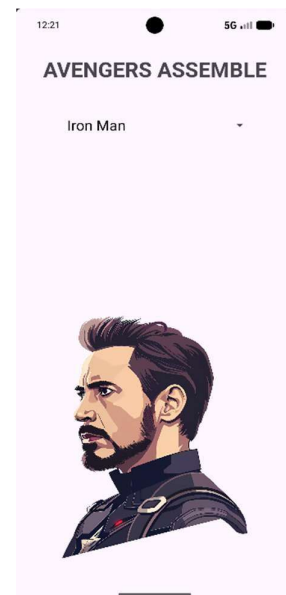
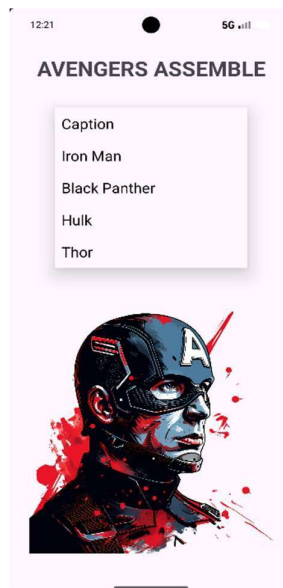
        ArrayAdapter<CharSequence> adapter = ArrayAdapter.createFromResource(this,
            R.array.avengers,
                R.layout.avenger_text);
        spinner.setAdapter(adapter);

        spinner.setOnItemClickListener(new AdapterView.OnItemClickListener() {
            @Override
            public void onItemClick(AdapterView<?> parent, View view, int position, long id)
```

```
{  
    imageView.setImageResource(images[position]);  
}  
  
@Override  
public void onNothingSelected(AdapterView<?> parent) {  
  
}  
});
```

```
ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->  
{  
    Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());  
    v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);  
    return insets;  
});  
}
```

Output:



Practical-7

Aim: Create a discount calculator app.

Activity_main.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="DISCOUNT CALCULATOR"
        android:textSize="30sp"
        android:textAlignment="center"
        android:textStyle="bold"
        android:layout_marginTop="20dp"
    />

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:gravity="center"
    >
        <EditText
            android:id="@+id/amount"
            android:layout_width="300dp"
            android:layout_height="wrap_content"
            android:hint="Enter Bill Amount"
            android:textSize="28dp"
            android:textAlignment="center"
            android:layout_marginBottom="20dp"
            android:inputType="numberDecimal"/>

        <TextView
            android:id="@+id/result"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:textSize="28sp"
            android:padding="10dp"
            android:layout_marginBottom="20dp"
```

```
<RadioGroup
    android:id="@+id/radioGroup"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:layout_marginTop="20dp">

    <RadioButton
        android:id="@+id/rb10"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20dp"
        android:text="10% Discount"/>

    <RadioButton
        android:id="@+id/rb15"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20dp"
        android:text="15% Discount"/>

    <RadioButton
        android:id="@+id/rb20"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20dp"
        android:text="20% Discount"/>
</RadioGroup>

<Button
    android:id="@+id/btn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Calculate"
    android:textSize="28dp"
    android:backgroundTint="@color/black"
    android:layout_marginTop="20dp"/>

</LinearLayout>
</LinearLayout>
```

Colors.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFF</color>
```

</resources>

Strings.xml code:

```
<resources>
    <string name="app_name">Discount_calc</string>
</resources>
```

MainActivity.java code:

```
package com.example.discount_calc;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        EditText amount;
        TextView result;
        RadioGroup radioGroup;
        Button btn;

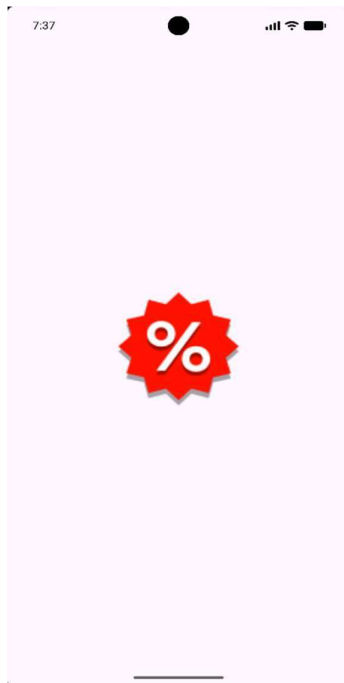
        amount = findViewById(R.id.amount);
        result = findViewById(R.id.result);
        radioGroup = findViewById(R.id.radioGroup);
        btn = findViewById(R.id.btn);

        btn.setOnClickListener(new View.OnClickListener() {
            @Override
```

```
public void onClick(View v) {  
  
    int value = Integer.parseInt(amount.getText().toString());  
  
    int gotId = radioGroup.getCheckedRadioButtonId();  
    RadioButton selectedRadio = findViewById(gotId);  
  
    int discount = 0;  
  
    if(selectedRadio.getId() == R.id.rb10){  
        discount = 10;  
    } else if (selectedRadio.getId() == R.id.rb15) {  
        discount = 15;  
    } else if (selectedRadio.getId() == R.id.rb20) {  
        discount = 20;  
    }  
  
    int finalAmount = value - (value * discount/100);  
  
    result.setText("Discount: ₹" + finalAmount);  
  
    }  
});  
  
ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->  
{  
    Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());  
    v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);  
    return insets;  
});  
}
```



Output:



Practical-8

Aim: Create an app with a course selection RadioButton group.

Activity_main.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select Your Course"
        android:textSize="32sp"
        android:layout_marginTop="-100dp"
        android:textStyle="bold"/>

    <ImageView
        android:layout_width="200dp"
        android:layout_height="200dp"
        android:src="@drawable/boy"
    />

    <RadioGroup
        android:id="@+id/radioGroup"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp">

        <RadioButton
            android:id="@+id/rbBDA"
            android:textSize="20dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="BDA"/>

        <RadioButton
            android:id="@+id/rbCSE"
            android:textSize="20dp"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="CSE"/>

        <RadioButton
```

```
android:id="@+id/rbBCA"  
android:textSize="20dp"  
android:layout_width="wrap_content"  
android:layout_height="wrap_content"  
android:text="BCA"/>
```

```
<RadioButton  
    android:id="@+id/rbCSEAI"  
    android:textSize="20dp"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="CSE AI"/>  
</RadioGroup>
```

```
<TextView  
    android:id="@+id/tvTIC"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:textSize="18sp"  
    android:layout_marginTop="25dp"  
    android:textStyle="bold"  
    android:text="TIC will appear here"/>
```

```
</LinearLayout>
```

Colors.xml code:

```
<?xml version="1.0" encoding="utf-8"?>  
<resources>  
    <color name="black">#FF000000</color>  
    <color name="white">#FFFFFFFF</color>  
</resources>
```

Strings.xml code:

```
<resources>  
    <string name="app_name">Course_Selection</string>  
</resources>
```

MainActivity.java code:

```
package com.example.course_selection;  
  
import android.os.Bundle;  
import android.widget.RadioGroup;  
import android.widget.TextView;  
  
import androidx.activity.EdgeToEdge;
```



```
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        RadioGroup radioGroup;
        TextView tvTIC;

        radioGroup = findViewById(R.id.radioGroup);
        tvTIC = findViewById(R.id.tvTIC);

        radioGroup.setOnCheckedChangeListener((group, checkedId) -> {
            if (checkedId == R.id.rbBDA)
                tvTIC.setText("TIC: 130 Credits");
            else if (checkedId == R.id.rbCSE)
                tvTIC.setText("TIC: 140 Credits");
            else if (checkedId == R.id.rbBCA)
                tvTIC.setText("TIC: 120 Credits");
            else if (checkedId == R.id.rbCSEAI)
                tvTIC.setText("TIC: 145 Credits");
        });

    }
}
```




Parul™
University

Faculty of Engineering & Technology
Subject Name : Mobile App Development
Subject Code : 303105380
B.Tech CSE Year 3rd Semester 6th

Output:

Select Your Course



☒ BDA

☐ CSE

☐ BCA

☐ CSE AI

TIC: 130 Credits

Practical-9

Aim: Create a shopping list app using checkboxes.

Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/shopping_list"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_marginBottom="16dp"/>

    <CheckBox
        android:id="@+id/cb_milk"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/milk" />

    <CheckBox
        android:id="@+id/cb_bread"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/bread" />

    <CheckBox
        android:id="@+id/cb_eggs"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/eggs" />

    <CheckBox
        android:id="@+id/cb_apples"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/apples" />
```

```
<TextView
    android:id="@+id/tv_selected_items"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="24dp"
    android:text="@string/selected_items_prefix"
    android:textSize="18sp" />
```

```
</LinearLayout>
```

Colors.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFF</color>
</resources>
```

Strings.xml:

```
<resources>
    <string name="app_name">Project_9</string>
    <string name="shopping_list">Shopping List</string>
    <string name="milk">Milk</string>
    <string name="bread">Bread</string>
    <string name="eggs">Eggs</string>
    <string name="apples">Apples</string>
    <string name="selected_items_prefix">Selected items: </string>
    <string name="none">None</string>
</resources>
```

MainActivity.java:

```
package com.example.project_9;

import android.os.Bundle;
import android.widget.CheckBox;
import android.widget.CompoundButton;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
```



```
import java.util.ArrayList;
import java.util.List;

public class MainActivity extends AppCompatActivity {

    private CheckBox cbMilk, cbBread, cbEggs, cbApples;
    private TextView tvSelectedItems;
    private List<CheckBox> checkBoxList;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
        {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });

        cbMilk = findViewById(R.id.cb_milk);
        cbBread = findViewById(R.id.cb_bread);
        cbEggs = findViewById(R.id.cb_eggs);
        cbApples = findViewById(R.id.cb_apples);
        tvSelectedItems = findViewById(R.id.tv_selected_items);

        checkBoxList = new ArrayList<>();
        checkBoxList.add(cbMilk);
        checkBoxList.add(cbBread);
        checkBoxList.add(cbEggs);
        checkBoxList.add(cbApples);

        CompoundButton.OnCheckedChangeListener changeListener = (buttonView,
        isChecked) -> updateSelectedItems();

        for (CheckBox cb : checkBoxList) {
            cb.setOnCheckedChangeListener(changeListener);
        }

        // Initial update
        updateSelectedItems();
    }

    private void updateSelectedItems() {
        StringBuilder selectedItems = new
```



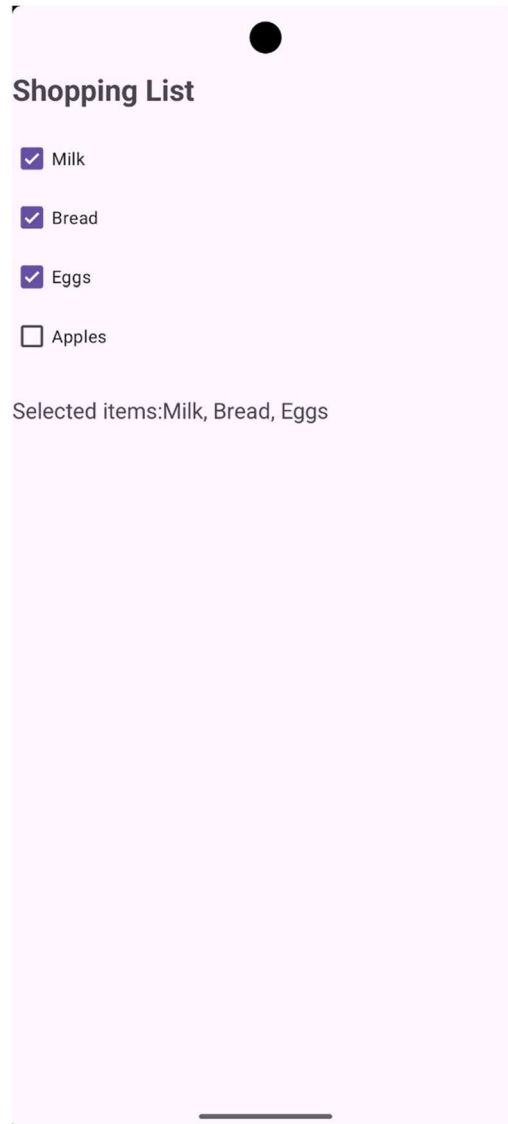
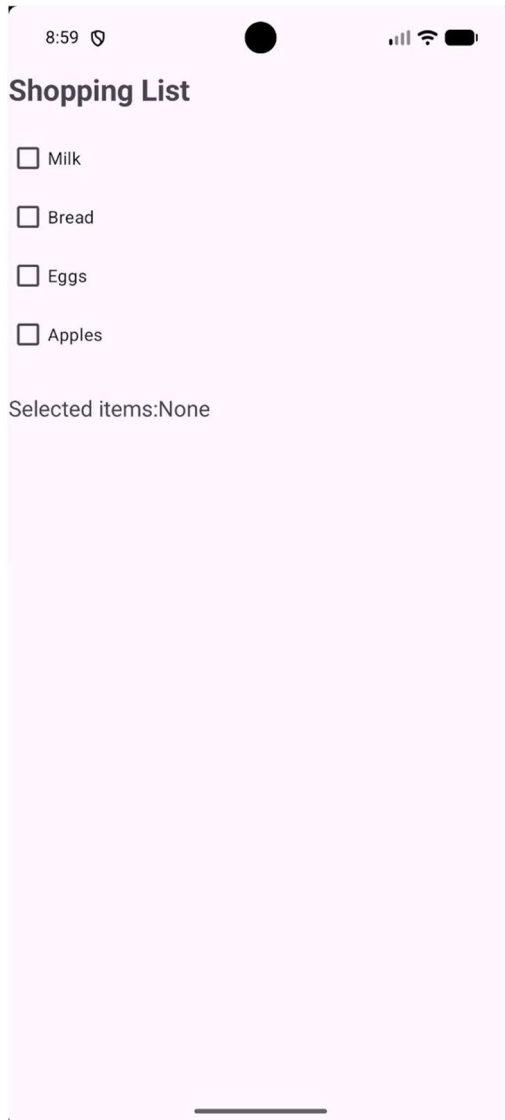
```
StringBuilder(getString(R.string.selected_items_prefix));
boolean first = true;
int count = 0;

for (CheckBox cb : checkBoxList) {
    if (cb.isChecked()) {
        if (!first) {
            selectedItems.append(", ");
        }
        selectedItems.append(cb.getText().toString());
        first = false;
        count++;
    }
}

if (count == 0) {
    tvSelectedItems.setText(getString(R.string.selected_items_prefix) +
    getString(R.string.none));
} else {
    tvSelectedItems.setText(selectedItems.toString());
}
}
```



Output:



Practical-10

Aim: Create a login and registration app

Activity_home.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <ImageView
        android:layout_width="match_parent"
        android:layout_height="250dp"
        android:layout_marginTop="-100dp"
        android:layout_marginBottom="100dp"
        android:src="@drawable/user"/>

    <TextView
        android:id="@+id/tvWelcome"
        android:text="Welcome User"
        android:textSize="32sp"
        android:textStyle="bold"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/logoutBtn"
        android:backgroundTint="#ffbc3e"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="24dp"
        android:text="Logout"/>

</LinearLayout>
```

Activity_login.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="30dp"
    android:gravity="center">
```

```
android:layout_width="match_parent"  
android:layout_height="match_parent">
```

```
<ImageView  
    android:layout_width="match_parent"  
    android:layout_height="250dp"  
    android:layout_marginTop="-100dp"  
    android:layout_marginBottom="100dp"  
    android:src="@drawable/login"/>
```

```
<TextView  
    android:layout_width="match_parent"  
    android:layout_height="50dp"  
    android:text="Login Here"  
    android:textAlignment="center"  
    android:textSize="30dp"  
    android:textStyle="bold"  
>
```

```
<EditText  
    android:id="@+id/etLoginUsername"  
    android:hint="Username"  
    android:textSize="24dp"  
    android:height="48dp"  
    android:inputType="text"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

```
<EditText  
    android:id="@+id/etLoginPassword"  
    android:hint="Password"  
    android:textSize="24dp"  
    android:height="48dp"  
    android:inputType="textPassword"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

```
<Button  
    android:id="@+id/btnLogin"  
    android:text="Login"  
    android:textSize="24dp"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

```
</LinearLayout>
```

Activity_register.xml:

```
<?xml version="1.0" encoding="utf-8"?>  
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```




```
android:orientation="vertical"  
android:padding="30dp"  
android:gravity="center"  
android:layout_width="match_parent"  
android:layout_height="match_parent">
```

```
<ImageView  
    android:layout_width="match_parent"  
    android:layout_height="250dp"  
    android:layout_marginTop="-100dp"  
    android:layout_marginBottom="100dp"  
    android:src="@drawable/reg"/>
```

```
<TextView  
    android:layout_width="match_parent"  
    android:layout_height="50dp"  
    android:text="Register Here"  
    android:textAlignment="center"  
    android:textSize="30dp"  
    android:textStyle="bold"  
>
```

```
<EditText  
    android:id="@+id/etUsername"  
    android:hint="Username"  
    android:textSize="24dp"  
    android:height="48dp"  
    android:inputType="text"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

```
<EditText  
    android:id="@+id/etPassword"  
    android:hint="Password"  
    android:textSize="24dp"  
    android:height="48dp"  
    android:inputType="textPassword"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

```
<Button  
    android:id="@+id/btnRegister"  
    android:text="Register"  
    android:textSize="24dp"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

```
<TextView  
    android:id="@+id/tvLogin"  
    android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="Already Registered? Login"
        android:textSize="16sp"
        android:textColor="@android:color/holo_blue_dark"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

Colors.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
</resources>
```

Strings.xml:

```
<resources>
    <string name="app_name">LoginApp</string>
</resources>
```

MainActivity.java :

```
package com.example.loginapp;

import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_register);

        EditText username = findViewById(R.id.etUsername);
```



```
EditText password = findViewById(R.id.etPassword);

Button btnRegister;
btnRegister = findViewById(R.id.btnRegister);

TextView tvLogin = findViewById(R.id.tvLogin);

tvLogin.setOnClickListener(v -> {
    Intent intent = new Intent(MainActivity.this, LoginActivity.class);
    startActivity(intent);
});

btnRegister.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        String user = username.getText().toString();
        String pass = password.getText().toString();

        SharedPreferences sp = getSharedPreferences("UserData", MODE_PRIVATE);

        SharedPreferences.Editor editor = sp.edit();

        editor.putString("username", user);
        editor.putString("password", pass);
        editor.apply();

        Toast.makeText(MainActivity.this, "Registered Successfully",
        Toast.LENGTH_SHORT).show();

        // Redirect to Login Page
        Intent i = new Intent(MainActivity.this, LoginActivity.class);
        startActivity(i);
        finish();

    }
});

}
```

LoginActivity.java:

```
package com.example.loginapp;

import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
```



```
import android.view.View;
import android.widget.*;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;

public class LoginActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);

        EditText username, password;
        Button login;

        username = findViewById(R.id.etLoginUsername);
        password = findViewById(R.id.etLoginPassword);
        login = findViewById(R.id.btnLogin);

        SharedPreferences sp = getSharedPreferences("UserData", MODE_PRIVATE);

        boolean isLoggedIn = sp.getBoolean("isLoggedIn", false);

        if(isLoggedIn){
            startActivity(new Intent(LoginActivity.this, activity_home.class));
            finish();
        }

        login.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                String user = username.getText().toString();
                String pass = password.getText().toString();

                String savedUser = sp.getString("username", "");
                String savedPass = sp.getString("password", "");

                if(user.equals(savedUser) && pass.equals(savedPass)){

                    SharedPreferences.Editor editor = sp.edit();
                    editor.putBoolean("isLoggedIn", true);
                    editor.apply();

                    startActivity(new Intent(LoginActivity.this, activity_home.class));
```

```
        finish();

    }else{
        Toast.makeText(LoginActivity.this,"Invalid Username or
Password",Toast.LENGTH_SHORT).show();
    }
}
});
}
```

Activity_home.java:

```
package com.example.loginapp;

import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class activity_home extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_home);

        Button logout;
        logout = findViewById(R.id.logoutBtn);

        TextView tvWelcome;
        tvWelcome = findViewById(R.id.tvWelcome);

        SharedPreferences sp = getSharedPreferences("UserData", MODE_PRIVATE);
        String username = sp.getString("username", "User");

        tvWelcome.setText("Welcome " + username);
    }
}
```



```
AlertDialog.Builder builder = new
AlertDialog.Builder(activity_home.this);
builder.setTitle("Success");
builder.setMessage("Welcome " + username);
builder.setPositiveButton("OK", null);
builder.show();

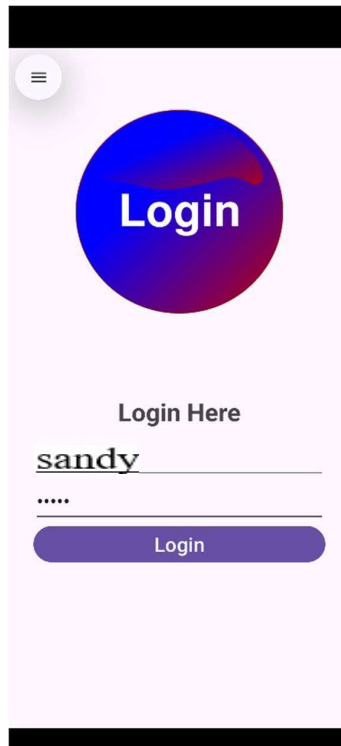
logout.setOnClickListener(v -> {

    // for simple logout button ---

    SharedPreferences.Editor editor = sp.edit();
    editor.putBoolean("isLoggedIn", false);
    editor.apply();
    startActivity(new Intent(activity_home.this, LoginActivity.class));
    finish();

});
}
```

Output:



Practical-11

Aim: Create a login and registration app

Activity_home.xml code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <ImageView
        android:layout_width="match_parent"
        android:layout_height="250dp"
        android:layout_marginTop="-100dp"
        android:layout_marginBottom="100dp"
        android:src="@drawable/user"/>

    <TextView
        android:id="@+id/tvWelcome"
        android:text="Welcome User"
        android:textSize="32sp"
        android:textStyle="bold"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/logoutBtn"
        android:backgroundTint="#ffbc3e"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="24dp"
        android:text="Logout"/>

</LinearLayout>
```

Activity_login.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="30dp"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
```



```
<ImageView  
    android:layout_width="match_parent"  
    android:layout_height="250dp"  
    android:layout_marginTop="-100dp"  
    android:layout_marginBottom="100dp"  
    android:src="@drawable/login"/>
```

```
<TextView  
    android:layout_width="match_parent"  
    android:layout_height="50dp"  
    android:text="Login Here"  
    android:textAlignment="center"  
    android:textSize="30dp"  
    android:textStyle="bold"  
>
```

```
<EditText  
    android:id="@+id/etLoginUsername"  
    android:hint="Username"  
    android:textSize="24dp"  
    android:height="48dp"  
    android:inputType="text"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

```
<EditText  
    android:id="@+id/etLoginPassword"  
    android:hint="Password"  
    android:textSize="24dp"  
    android:height="48dp"  
    android:inputType="textPassword"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

```
<Button  
    android:id="@+id/btnLogin"  
    android:text="Login"  
    android:textSize="24dp"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>  
</LinearLayout>
```

Activity_register.xml:

```
<?xml version="1.0" encoding="utf-8"?>  
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"  
    android:orientation="vertical"  
    android:padding="30dp"
```

```
android:gravity="center"  
android:layout_width="match_parent"  
android:layout_height="match_parent">
```

```
<ImageView  
    android:layout_width="match_parent"  
    android:layout_height="250dp"  
    android:layout_marginTop="-100dp"  
    android:layout_marginBottom="100dp"  
    android:src="@drawable/reg"/>
```

```
<TextView  
    android:layout_width="match_parent"  
    android:layout_height="50dp"  
    android:text="Register Here"  
    android:textAlignment="center"  
    android:textSize="30dp"  
    android:textStyle="bold"  
>
```

```
<EditText  
    android:id="@+id/etUsername"  
    android:hint="Username"  
    android:textSize="24dp"  
    android:height="48dp"  
    android:inputType="text"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

```
<EditText  
    android:id="@+id/etPassword"  
    android:hint="Password"  
    android:textSize="24dp"  
    android:height="48dp"  
    android:inputType="textPassword"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

```
<Button  
    android:id="@+id/btnRegister"  
    android:text="Register"  
    android:textSize="24dp"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

```
<TextView  
    android:id="@+id/tvLogin"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="Already Registered? Login"
```



```
        android:textSize="16sp"
        android:textColor="@android:color/holo_blue_dark"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

Colors.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFF</color>
</resources>
```

Strings.xml:

```
<resources>
    <string name="app_name">LoginApp</string>
</resources>
```

MainActivity.java :

```
package com.example.loginapp;

import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_register);

        EditText username = findViewById(R.id.etUsername);
        EditText password = findViewById(R.id.etPassword);
```

```
Button btnRegister;  
btnRegister = findViewById(R.id.btnRegister);  
  
TextView tvLogin = findViewById(R.id.tvLogin);  
  
tvLogin.setOnClickListener(v -> {  
    Intent intent = new Intent(MainActivity.this, LoginActivity.class);  
    startActivity(intent);  
});  
  
btnRegister.setOnClickListener(new View.OnClickListener() {  
    @Override  
    public void onClick(View v) {  
  
        String user = username.getText().toString();  
        String pass = password.getText().toString();  
  
        SharedPreferences sp = getSharedPreferences("UserData", MODE_PRIVATE);  
  
        SharedPreferences.Editor editor = sp.edit();  
  
        editor.putString("username", user);  
        editor.putString("password", pass);  
        editor.apply();  
  
        Toast.makeText(MainActivity.this, "Registered Successfully",  
        Toast.LENGTH_SHORT).show();  
  
        // Redirect to Login Page  
        Intent i = new Intent(MainActivity.this, LoginActivity.class);  
        startActivity(i);  
        finish();  
  
    }  
});  
}
```

LoginActivity.java:

```
package com.example.loginapp;  
  
import android.content.Intent;  
import android.content.SharedPreferences;  
import android.os.Bundle;  
import android.view.View;  
import android.widget.*;
```

```
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;

public class LoginActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_login);

        EditText username, password;
        Button login;

        username = findViewById(R.id.etLoginUsername);
        password = findViewById(R.id.etLoginPassword);
        login = findViewById(R.id.btnLogin);

        SharedPreferences sp = getSharedPreferences("UserData", MODE_PRIVATE);

        boolean isLoggedIn = sp.getBoolean("isLoggedIn", false);

        if(isLoggedIn){
            startActivity(new Intent(LoginActivity.this, activity_home.class));
            finish();
        }

        login.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                String user = username.getText().toString();
                String pass = password.getText().toString();

                String savedUser = sp.getString("username", "");
                String savedPass = sp.getString("password", "");

                if(user.equals(savedUser) && pass.equals(savedPass)){

                    SharedPreferences.Editor editor = sp.edit();
                    editor.putBoolean("isLoggedIn", true);
                    editor.apply();

                    startActivity(new Intent(LoginActivity.this, activity_home.class));
                    finish();
                }
            }
        });
    }
}
```

```
        }else{
            Toast.makeText(LoginActivity.this,"Invalid
Username or Password",Toast.LENGTH_SHORT).show();
        }
    }
});
}
```

Activity_home.java:

```
package com.example.loginapp;

import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class activity_home extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_home);

        Button logout;
        logout = findViewById(R.id.logoutBtn);

        TextView tvWelcome;
        tvWelcome = findViewById(R.id.tvWelcome);

        SharedPreferences sp = getSharedPreferences("UserData", MODE_PRIVATE);
        String username = sp.getString("username", "User");

        tvWelcome.setText("Welcome " + username);

        AlertDialog.Builder builder = new AlertDialog.Builder(activity_home.this);
        builder.setTitle("Success");
```



```
builder.setMessage("Welcome " + username);  
builder.setPositiveButton("OK", null);  
builder.show();
```

```
logout.setOnClickListener(v -> {
```

```
// for showing popup on logout button ----
```

```
builder.setTitle("Logout");  
builder.setMessage("Are you sure you want to logout?");
```

```
builder.setPositiveButton("OK", (dialog, which) -> {
```

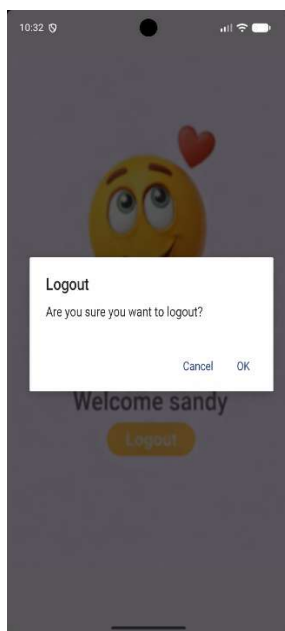
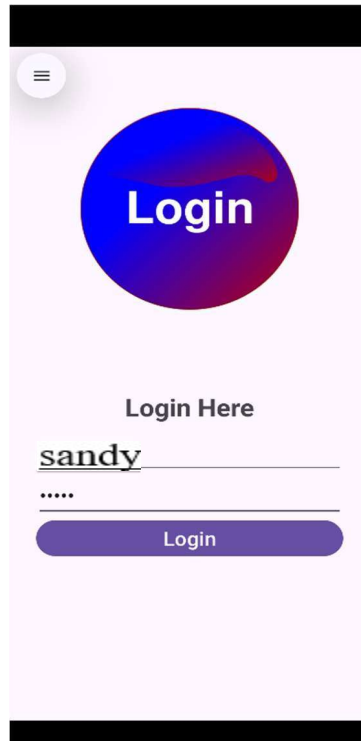
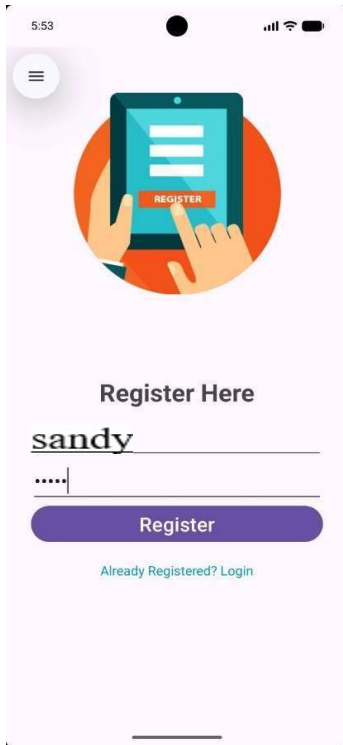
```
    SharedPreferences.Editor editor = sp.edit();  
    editor.putBoolean("isLoggedIn", false);  
    editor.apply();  
    startActivity(new Intent(activity_home.this, LoginActivity.class));  
    finish();  
});
```

```
builder.setNegativeButton("Cancel", null);  
builder.show();
```

```
});
```

```
}  
}
```

output:



Practical-12

Aim: Create an app with a menu

Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:id="@+id/tv"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select Option From Menu"
        android:textStyle="bold"
        android:textSize="22sp" />

</LinearLayout>
```

Simple_menu.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item android:id="@+id/op1" android:title="Apple"/>
    <item android:id="@+id/op2" android:title="Grapes"/>
    <item android:id="@+id/op3" android:title="Pineapple"/>
    <item android:id="@+id/op4" android:title="Oranges"/>
    <item android:id="@+id/op5" android:title="Watermelon"/>
</menu>
```

Colors.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFF</color>
</resources>
```

Strings.xml:

```
<resources>
    <string name="app_name">Project-12</string>
```

</resources>

MainActivity.java:

```
package com.example.project_12;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    TextView tv;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        tv = findViewById(R.id.tv);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {

        getMenuInflater().inflate(R.menu.simple_menu, menu);

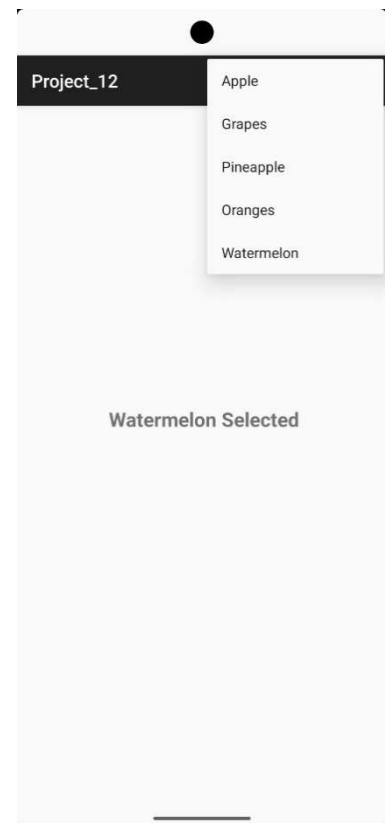
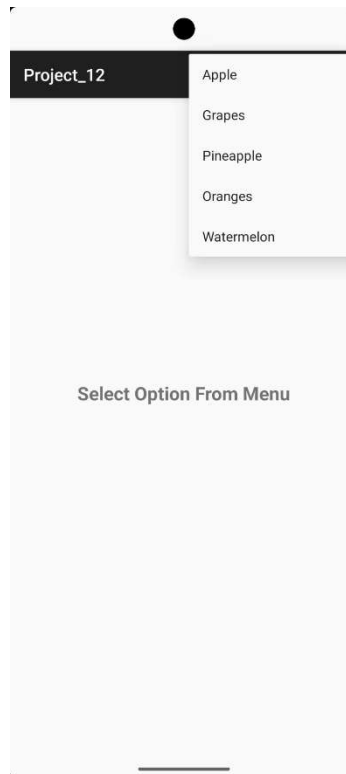
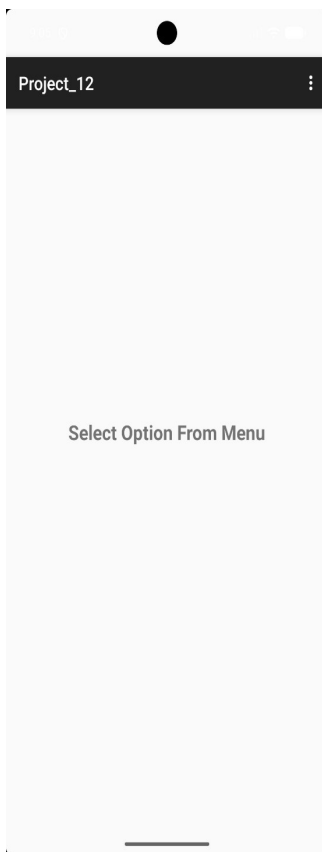
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(MenuItem item) {

        if(item.getItemId() == R.id.op1){
            tv.setText(" Apple Selected");
        }
        else if(item.getItemId() == R.id.op2){
            tv.setText(" Grapes Selected");
        }
        else if(item.getItemId() == R.id.op3){
            tv.setText(" Pineapple Selected");
        }
    }
}
```

```
else if(item.getItemId() == R.id.op4){  
    tv.setText(" Oranges Selected");  
}  
else if(item.getItemId() == R.id.op5){  
    tv.setText(" Watermelon Selected");  
}  
  
return true;  
}  
}
```

Output:



Practical-13

Aim: Build an app using LinearLayout.

Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    android:gravity="center"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:textAlignment="center"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="hello world"
        android:textSize="18sp"
        android:padding="8dp" />

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Convert to Uppercase"
        android:layout_gravity="center"
        android:layout_marginTop="20dp" />

    <EditText
        android:id="@+id/editText"
        android:layout_width="250dp"
        android:layout_gravity="center"
        android:layout_height="48dp"
        android:hint="Result will appear here"
        android:layout_marginTop="16dp"
        android:inputType="text" />

    <TextView
```

```
android:id="@+id/keyEventStatus"  
android:layout_width="match_parent"  
android:layout_height="wrap_content"  
android:layout_marginTop="16dp"  
android:textAlignment="center"  
android:text="Type in EditText to see live updates"  
android:textColor="@android:color/darker_gray" />
```

```
</LinearLayout>
```

Colors.xml:

```
<?xml version="1.0" encoding="utf-8"?>  
<resources>  
    <color name="black">#FF000000</color>  
    <color name="white">#FFFFFFF</color>  
</resources>
```

Strings.xml:

```
<resources>  
    <string name="app_name">Project_13</string>  
</resources>
```

MainActivity.java:

```
package com.example.project_13;  
  
import android.os.Bundle;  
import android.text.Editable;  
import android.text.TextWatcher;  
import android.widget.Button;  
import android.widget.EditText;  
import android.widget.TextView;  
  
import androidx.activity.EdgeToEdge;  
import androidx.appcompat.app.AppCompatActivity;  
import androidx.core.graphics.Insets;  
import androidx.core.view.ViewCompat;  
import androidx.core.view.WindowInsetsCompat;  
  
public class MainActivity extends AppCompatActivity {  
  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        EdgeToEdge.enable(this);  
        setContentView(R.layout.activity_main);  
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
```

```
{
    Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
    return insets;
});

TextView textView = findViewById(R.id.textView);
Button button = findViewById(R.id.button);
EditText editText = findViewById(R.id.editText);
TextView keyEventStatus = findViewById(R.id.keyEventStatus);

// Convert TextView content to uppercase and display in EditText on button click
button.setOnClickListener(v -> {
    String originalText = textView.getText().toString();
    String upperText = originalText.toUpperCase();
    editText.setText(upperText);
});

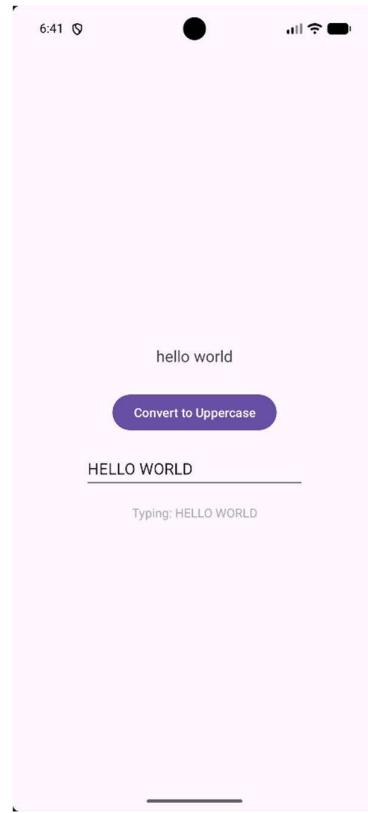
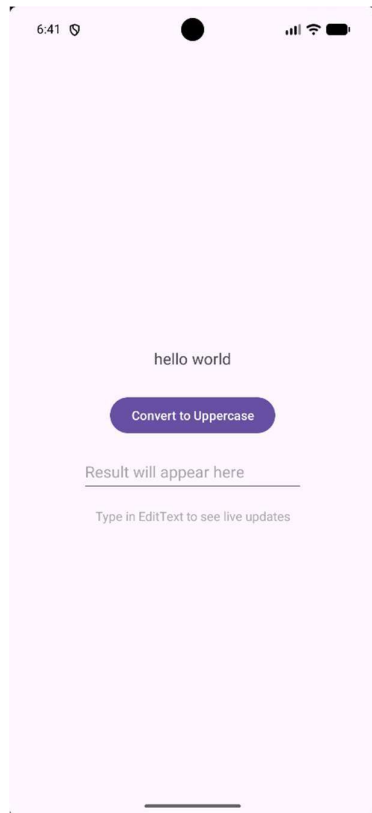
// Respond to key events in EditText without button press
editText.addTextChangedListener(new TextWatcher() {
    @Override
    public void beforeTextChanged(CharSequence s, int start, int count, int after) {
    }

    @Override
    public void onTextChanged(CharSequence s, int start, int before, int count) {
        // Update live status as the user types
        keyEventStatus.setText("Typing: " + s.toString());
    }

    @Override
    public void afterTextChanged(Editable s) {
    }
});
}
```



Output:



Practical-14

Aim: Create an app with **TableLayout** and custom styles.

Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<TableLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp"
    android:stretchColumns="1"
    android:gravity="center"
    tools:context=".MainActivity">

    <TableRow>
        <TextView
            style="@style/CustomTextViewStyle"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Name:" />

        <EditText
            android:id="@+id/editTextName"
            android:layout_width="match_parent"
            android:layout_height="48dp"
            android:hint="Enter your name" />
    </TableRow>

    <TableRow android:layout_marginTop="16dp">
        <Button
            android:id="@+id/buttonSubmit"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Submit" />

        <Button
            android:id="@+id/buttonClear"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Clear" />
    </TableRow>

</TableLayout>
```




Colors.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFF</color>
</resources>
```

Strings.xml:

```
<resources>
    <string name="app_name">Project_14</string>
</resources>
```

Styles.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <style name="CustomTextViewStyle">
        <item name="android:textSize">18sp</item>
        <item name="android:textColor">#3F51B5</item>
        <item name="android:textStyle">bold</item>
        <item name="android:padding">8dp</item>
    </style>
</resources>
```

MainActivity.java:

```
package com.example.project_14;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
```

```
setContentView(R.layout.activity_main);

ViewPager.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
{
    Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
    return insets;
});

EditText editTextName = findViewById(R.id.editTextName);
Button buttonSubmit = findViewById(R.id.buttonSubmit);
Button buttonClear = findViewById(R.id.buttonClear);

buttonSubmit.setOnClickListener(v -> {
    String name = editTextName.getText().toString();
    if (!name.isEmpty()) {
        Toast.makeText(MainActivity.this, "Hello, " + name + "!",
Toast.LENGTH_SHORT).show();
    } else {
        Toast.makeText(MainActivity.this, "Please enter a name",
Toast.LENGTH_SHORT).show();
    }
});

buttonClear.setOnClickListener(v -> editTextName.setText(""));
}
```



Output:

6:58

Name:

Submit Clear

7:35

Name:

Submit Clear

Practical-15

Aim: Create an app with SQLite database operations.

Activity_main.xml:

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:orientation="vertical"
    android:padding="24dp"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#FFF3E0">

    <ImageView
        android:layout_width="match_parent"
        android:layout_height="200dp"
        android:src="@drawable/todo"/>

    <TextView
        android:text="APPLICATION"
        android:textSize="28sp"
        android:textStyle="bold"
        android:textColor="#E65100"
        android:layout_marginBottom="30dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/taskInput"
        android:hint="Enter Task"
        android:textSize="24dp"
        android:textAlignment="center"
        android:height="48dp"
        android:layout_width="220dp"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/addBtn"
        android:text="Add Task"
        android:backgroundTint="#FB8C00"
        android:textColor="#FFFFFF"
        android:textSize="20sp"
        android:layout_marginTop="20dp"
        android:layout_width="220dp"
```

```
android:layout_height="wrap_content"/>
```

```
<Button
    android:id="@+id/viewBtn"
    android:text="View Tasks"
    android:backgroundTint="#EF6C00"
    android:textColor="#FFFFFF"
    android:layout_marginTop="10dp"
    android:textSize="20sp"
    android:layout_width="220dp"
    android:layout_height="wrap_content"/>
```

```
</LinearLayout>
```

Activity_view_task.xml:

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:orientation="vertical"
    android:padding="20dp"
    android:background="#FFF3E0"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
```

```
<TextView
    android:text="Your Tasks"
    android:textSize="24sp"
    android:textStyle="bold"
    android:textColor="#E65100"
    android:layout_marginTop="20dp"
    android:layout_marginBottom="20dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>
```

```
<ListView
    android:id="@+id/listView"
    android:layout_width="match_parent"
    android:layout_height="match_parent"/>
```

```
</LinearLayout>
```

Colors.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
</resources>
```

Strings.xml:

```
<resources>
    <string name="app_name">Todo</string>
</resources>
```

MainActivity.java:

```
package com.example.todo;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    DatabaseHelper db;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        EditText taskInput;
        Button addBtn, viewBtn;

        taskInput = findViewById(R.id.taskInput);
        addBtn = findViewById(R.id.addBtn);
        viewBtn = findViewById(R.id.viewBtn);

        db = new DatabaseHelper(this);

        // ADD TASK BUTTON
        addBtn.setOnClickListener(new View.OnClickListener() {
            @Override
```

```
public void onClick(View v) {

    String task = taskInput.getText().toString();

    if(task.isEmpty()){
        Toast.makeText(MainActivity.this,"Enter
Task",Toast.LENGTH_SHORT).show();
    }else{

        boolean result = db.insertTask(task);

        if(result){
            Toast.makeText(MainActivity.this,"Task
Added",Toast.LENGTH_SHORT).show();
            taskInput.setText("");
        }else{
            Toast.makeText(MainActivity.this,"Insert
Failed",Toast.LENGTH_SHORT).show();
        }
    }
}

// VIEW TASK BUTTON (open second page)
viewBtn.setOnClickListener(v -> {
    startActivity(new Intent(MainActivity.this, activity_view_task.class));
});

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) ->
{
    Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
    return insets;
});
}
```

DatabaseHelper.java:

```
package com.example.todo;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
```




```
public class DatabaseHelper extends SQLiteOpenHelper {

    // Constructor - creates database named TodoDB
    public DatabaseHelper(Context context) {
        super(context, "TodoDB", null, 1);
    }

    // This runs when database is created for the first time
    @Override
    public void onCreate(SQLiteDatabase db) {

        // Create a table named tasks
        // id will increase automatically
        // task will store text
        db.execSQL("CREATE TABLE tasks(id INTEGER PRIMARY KEY
        AUTOINCREMENT, task TEXT)");
    }

    // Runs if database version changes
    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {

        // Delete old table
        db.execSQL("DROP TABLE IF EXISTS tasks");

        // Create table again
        onCreate(db);
    }

    // INSERT TASK
    public boolean insertTask(String task) {

        // open database for writing
        SQLiteDatabase db = this.getWritableDatabase();

        // object to store column values
        ContentValues values = new ContentValues();

        // column name , value
        values.put("task", task);

        // insert into table
        long result = db.insert("tasks", null, values);

        // check if inserted
        if(result == -1){
            return false;
        }else{
            return true;
        }
    }
}
```



```
}  
}  
  
// READ ALL TASKS  
public Cursor getTasks(){  
  
    SQLiteDatabase db = this.getReadableDatabase();  
  
    return db.rawQuery("SELECT * FROM tasks", null);  
}  
  
// UPDATE TASK  
public boolean updateTask(int id, String task){  
  
    SQLiteDatabase db = this.getWritableDatabase();  
  
    ContentValues values = new ContentValues();  
  
    values.put("task", task);  
  
    int result = db.update("tasks", values, "id=?", new String[]{String.valueOf(id)});  
  
    return result > 0;  
}  
  
// DELETE TASK  
public boolean deleteTask(int id){  
  
    SQLiteDatabase db = this.getWritableDatabase();  
  
    int result = db.delete("tasks", "id=?", new String[]{String.valueOf(id)});  
  
    return result > 0;  
}  
}
```

Activity_view_task.java:

```
package com.example.todo;  
  
import android.content.ContentValues;  
import android.content.Context;  
import android.database.Cursor;  
import android.database.sqlite.SQLiteDatabase;  
import android.database.sqlite.SQLiteOpenHelper;  
  
public class DatabaseHelper extends SQLiteOpenHelper {
```



```
// Constructor - creates database named TodoDB
public DatabaseHelper(Context context) {
    super(context, "TodoDB", null, 1);
}

// This runs when database is created for the first time
@Override
public void onCreate(SQLiteDatabase db) {

    // Create a table named tasks
    // id will increase automatically
    // task will store text
    db.execSQL("CREATE TABLE tasks(id INTEGER PRIMARY KEY
AUTOINCREMENT, task TEXT)");
}

// Runs if database version changes
@Override
public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {

    // Delete old table
    db.execSQL("DROP TABLE IF EXISTS tasks");

    // Create table again
    onCreate(db);
}

// INSERT TASK
public boolean insertTask(String task) {

    // open database for writing
    SQLiteDatabase db = this.getWritableDatabase();

    // object to store column values
    ContentValues values = new ContentValues();

    // column name , value
    values.put("task", task);

    // insert into table
    long result = db.insert("tasks", null, values);

    // check if inserted
    if(result == -1){
        return false;
    }else{
        return true;
    }
}
```



```
// READ ALL TASKS
public Cursor getTasks(){

    SQLiteDatabase db = this.getReadableDatabase();

    return db.rawQuery("SELECT * FROM tasks", null);
}

// UPDATE TASK
public boolean updateTask(int id, String task){

    SQLiteDatabase db = this.getWritableDatabase();

    ContentValues values = new ContentValues();

    values.put("task", task);

    int result = db.update("tasks", values, "id=?", new String[]{String.valueOf(id)});

    return result > 0;
}

// DELETE TASK
public boolean deleteTask(int id){

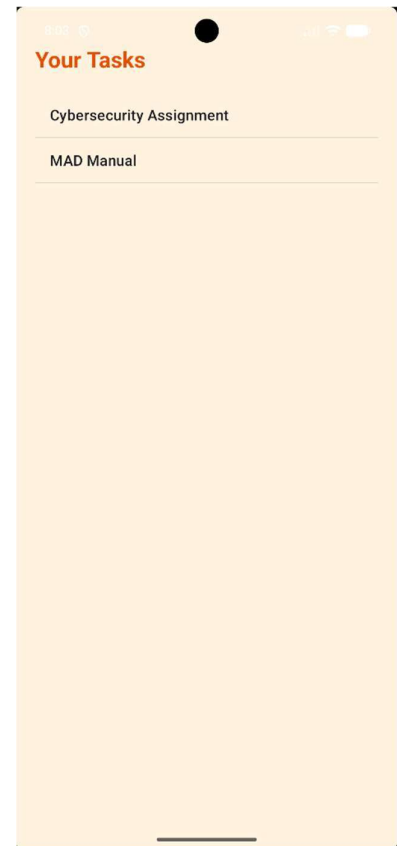
    SQLiteDatabase db = this.getWritableDatabase();

    int result = db.delete("tasks", "id=?", new String[]{String.valueOf(id)});

    return result > 0;
}
}
```



Output:



Practical-16

Aim: Create an app with three vertically aligned buttons

Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/main"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".MainActivity">

    <Button
        android:id="@+id/btnRed"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Red"
        app:layout_constraintBottom_toTopOf="@+id/btnGreen"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_chainStyle="packed" />

    <Button
        android:id="@+id/btnGreen"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:layout_marginBottom="16dp"
        android:text="Green"
        app:layout_constraintBottom_toTopOf="@+id/btnBlue"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/btnRed" />

    <Button
        android:id="@+id/btnBlue"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Blue"
```

```
app:layout_constraintBottom_toBottomOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintHorizontal_bias="0.5"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toBottomOf="@+id/btnGreen" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

Colors.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFF</color>
</resources>
```

Strings.xml:

```
<resources>
    <string name="app_name">Project_16</string>
</resources>
```

MainActivity.java:

```
package com.example.project_16;

import android.graphics.Color;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
    }
}
```

```
View mainLayout = findViewById(R.id.main);  
Button btnRed = findViewById(R.id.btnRed);  
Button btnGreen = findViewById(R.id.btnGreen);  
Button btnBlue = findViewById(R.id.btnBlue);
```

```
ViewCompat.setOnApplyWindowInsetsListener(mainLayout, (v, insets) -> {  
    Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());  
    v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);  
    return insets;  
});
```

```
btnRed.setOnClickListener(v -> mainLayout.setBackgroundColor(Color.RED));  
btnGreen.setOnClickListener(v -> mainLayout.setBackgroundColor(Color.GREEN));  
btnBlue.setOnClickListener(v -> mainLayout.setBackgroundColor(Color.BLUE));  
}  
}
```

Output:

